



Towards sustainable food consumption: understanding household food waste drivers for improved waste management in urban Vietnam

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Abstract

Households are a major source of food waste, contributing to greenhouse gas emissions, resource inefficiencies, and food insecurity. While the behavioural drivers of food waste are increasingly studied, few distinguish between households that primarily waste raw ingredients and those that primarily waste cooked food—an important distinction for urban-level interventions. This study uses primary survey data from 642 households in Ho Chi Minh City, Vietnam. It applies Covariance-Based Structural Equation Modelling (CB-SEM) to investigate the psychological and behavioural factors shaping food waste during home cooking. Findings reveal that intention does not significantly translate into reduced waste, indicating an intention–behaviour gap. Cooking ability and understanding of family food preferences consistently reduce food waste across household types, whereas planning and storage practices show limited effects. Notably, feelings of guilt are unexpectedly associated with higher waste among cooked-food-wasting households—potentially due to post-rationalisation. A supplementary exploratory Variance-Based SEM (VB-SEM) model of 113 zero-waste households shows that intention more strongly predicts this group’s planning and storage behaviours. These findings highlight the need for context-specific, behaviourally informed interventions that target distinct household segments. Differentiated strategies are essential to reducing food waste and promoting sustainable consumption in rapidly urbanising cities like Ho Chi Minh City.

Keywords Household food waste · Sustainable urban food consumption · Behavioural drivers · Ingredient vs cooked food waste · Structural equation modelling (SEM)

Introduction

Food waste is a critical component of urban solid waste streams, significantly contributes to greenhouse gas emissions, wastes natural and labour resources, and threatens

global food security [1]. In 2022, households worldwide were estimated to have wasted over one billion meals per day (equivalent to 376 billion meals annually) highlighting the persistent scale of the issue despite global prevention efforts [2]. Globally, approximately one-third of all food produced is either lost or wasted, with a significant share occurring at the consumption stage [3], underscoring the central role of households in advancing sustainable material cycles and waste management.

In rapidly urbanising environments, household food waste is a dominant source of organic waste, much of which originates from home cooking [4]. A growing body of research has investigated behavioural and psychological factors influencing household food waste, particularly attitudes, norms, and intentions shaping waste-related behaviours [5, 6]. More recent studies have shifted toward the design of concrete interventions and management strategies. For example, Freitas et al. [7] examine food waste-reducing digital platforms in Brazil and show how institutional,

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cultural, and organisational barriers constrain their effectiveness in middle-income contexts, while Zhang et al. [8] demonstrate how psychological and socio-cultural factors, such as collective memory, shape household intentions to reduce food waste and thereby inform the design of behaviourally grounded interventions. Other research has adopted a system-level and policy-oriented perspective, including case-based analyses of food sector waste management strategies across different regional contexts [9] and reviews of multi-criteria decision-making tools that support the selection of integrated biowaste management strategies from prevention through valorisation [10]. Importantly, food waste in home cooking arises at different points in the cooking cycle, and these differences are behaviourally meaningful. Ingredient waste, such as unused or spoiled items, often stems from over-purchasing or poor storage. Cooked food waste, such as leftovers from oversized meals or unconsumed dishes, may reflect poor portioning, limited cooking skills, or a mismatch with household food preferences. Treating food waste as a single behavioural outcome risk obscures the specific behaviours behind it and may limit the design of effective, targeted interventions for urban waste reduction.

While several studies have investigated raw ingredients and cooked food waste in households, few have directly compared households based on which type of waste is more dominant. Early work by Van Garde and Woodburn (1987) distinguished between preparation waste, such as vegetable peels and unused ingredients and post-consumption waste from cooked meals in U.S. households [6]. Similarly, Quested et al. (2012) and Schmidt & Matthies (2018) investigated specific behavioural pathways linked to different waste types in the UK and Germany, respectively [11, 12]. However, these studies did not explicitly compare households that primarily waste raw ingredients with those that waste more cooked meals. In addition, most existing work focuses on high-income countries, with limited empirical research having been conducted in Southeast Asia. In Vietnam, where urbanisation is accelerating, only a handful of studies have examined household food waste behaviours [13, 14], but none have focused on behavioural differences between household waste types. This gap is significant given that in Ho Chi Minh City (HCMC) alone, households generated around 13,000 tons of solid waste per day, of which 50–60% is organic waste [15].

This study addresses these gaps by analysing the psychological and behavioural drivers of household food waste during home cooking in HCMC, and by comparing ingredient-wasting and cooked-food-wasting households. Guided by the Theory of Planned Behaviour (TPB) and previous empirical findings, we applied Covariance-Based Structural Equation Modelling (CB-SEM) to test key hypotheses and further conducted exploratory Variance-Based

SEM (VB-SEM) on a subgroup of zero-waste households. This mixed-SEM approach provides new insights into the dynamics of food waste generation in households and their implications for waste prevention, material cycle efficiency, and urban environmental management.

This study contributes to global efforts aligned with the United Nations Sustainable Development Goals—particularly SDG 11 (Sustainable Cities and Communities) and SDG 12 (Responsible Consumption and Production). By differentiating between household waste types and identifying context-specific behavioural patterns, this study contributes to both behavioural science and applied waste management. The findings support the design of targeted strategies to reduce organic waste in cities, to develop solutions for material cycles and to advance sustainable urban waste systems.

Literature review

Ingredient waste and cooked food waste

Ingredient waste may arise during food preparation, the spoilage of raw materials, or the wasting of leftover ingredients. Several analyses have found that poor planning during food purchasing or preparation can lead to significant ingredient waste [16, 17] or highlight improper storage as a major cause of ingredient spoilage [18]. Additionally, low kitchen confidence or limited preparation skills have been linked to increased waste during the preparation phase [19]. Cooked food waste may share some of the same drivers, such as storage practices and planning behaviours. However, it can also stem from different causes. For example, burnt or over-cooked meals, poor taste, or food that fails to meet household preferences [20].

Existing literature reveals that most studies tend to treat household food waste as a single overarching concept or only briefly acknowledge the two types of waste along with their respective drivers. However, factors influencing household food waste should likely differ between households that predominantly waste raw ingredients and those that predominantly waste cooked food, given the fundamentally different practices and decision points through which these wastes arise. Accordingly, this study hypothesises that the structural paths linking constructs to food waste outcomes differ significantly between households that waste more raw ingredients and those that waste more cooked food (H_1). This proposition is empirically tested in the subsequent analysis.

Growing food waste in urban households also creates significant pressure on municipal waste systems, with implications for energy use, emissions, and sanitation. In

rapidly urbanising areas like Ho Chi Minh City, a better understanding of food waste behaviour at the household level can inform urban waste prevention programs and support broader sustainability transitions.

Conceptual framework

The Theory of Planned Behaviour (TPB) is one of the most widely used frameworks in food waste studies [21]. As an extension of the Theory of Reasoned Action (TRA), TPB subsumes the core assumptions of TRA (attitudes, subjective norms, and behavioural intention) while incorporating perceived behavioural control) and therefore provides a sufficient and theoretically coherent foundation for analysing food waste behaviours. However, several studies have highlighted its limitations in capturing the complexity of food waste behaviours. While TPB effectively explains intentions, it often fails to account for the intention-behaviour gap, where strong intentions do not always translate into actions [22]. Furthermore, TPB overlooks important contextual and systemic barriers, such as packaging sizes [12], and does not address emotional factors, such as feelings of guilt, which can strongly influence food waste behaviours [23]. To address these limitations, this study develops an extended conceptual framework by synthesising empirical evidence on household food waste and integrating practice-based and emotional mechanisms in everyday home cooking.

TPB suggests that individuals' intentions to engage in behaviour are shaped by their attitudes toward it, perceived social norms, and perceived control over the behaviour [24]. These intentions, in turn, are likely to predict actual behaviour, with stronger intentions increasing the likelihood of the behaviour being performed [24]. Consistent with this theoretical logic, the following hypothesis is proposed and empirically tested:

H₂ The greater the intention to reduce food waste, the smaller the amount of household food waste.

Beyond intention, food waste in urban home cooking is shaped by a range of everyday practices through which food is planned, purchased, stored, and handled after meals. Effective planning, such as preparing shopping lists or checking current food supplies, helps reduce over-purchasing and impulsive buying [25, 26]. Conversely, bulk purchasing has been associated with higher levels of food waste when households are unable to manage large quantities effectively [27, 28]. However, this risk can be mitigated through proper portioning and sealed storage [29]. Inadequate storage often leads to spoilage or forgotten items, especially when household food environments are cluttered or disorganised [30],

while reusing leftovers or creatively repurposing cooked food into new meals is a key strategy to prevent unnecessary waste [26]. Taken together, these findings suggest that everyday home cooking practices may play a direct role in shaping household food waste outcomes. Therefore, this study proposes:

H₃ Home cooking practices—including food planning, purchasing, storage, and leftover handling—are significantly associated with household food waste.

From a psychological perspective, emotions such as guilt have been shown to influence food waste behaviour [23]. Individuals who perceive food waste as morally wrong or socially unacceptable tend to report lower levels of waste [31]. In some cultural contexts, finishing all prepared food is viewed as a moral obligation [31]. In Vietnam, where Buddhist values emphasise moderation and avoidance of waste, guilt may play a particularly salient role in shaping post-consumption food management decisions. Unlike attitudes or subjective norms, guilt is conceptualised here as a post-action affective response that influences subsequent food management decisions rather than intention formation. Accordingly, this study hypothesises:

H₄ Greater guilt is associated with lower food waste.

In addition to emotional factors, cooking ability is conceptualised as a practice-based capability that conditions whether perceived behavioural control can be effectively enacted in everyday cooking. While perceived behavioural control captures individuals' perceived ease or difficulty of performing a behaviour, cooking ability and understanding family food preferences represent enacted, practice-based capabilities that shape food waste outcomes during daily meal preparation. Skilled cooks are less likely to waste food, as they are better able to avoid cooking mistakes, manage ingredients efficiently, and creatively repurpose leftovers [20, 32]. Higher cooking ability, therefore, reduces the likelihood that food becomes unusable or discarded. This leads to the following hypothesis:

H₅ Greater cooking skills are linked to reduced household food waste.

Similarly, when individuals clearly understand their household's food needs, it results in more efficient home cooking. By recognising family members' preferences, portion sizes, and food likes, the primary cook can prepare flavourful meals with appropriate portions, reduce leftovers, and, ultimately, minimise household food waste [33]. Based on this reasoning, the study proposes:

H₆ A deeper understanding of family needs is linked to reduced household food waste.

The literature further suggests that demographic factors, such as education, age, and gender, significantly influence household food waste. Several studies have found that women are more likely to adopt sustainable food consumption behaviours than men [34, 35]; however, other research indicates that women may also generate more food waste than men [36]. Higher educational attainment is generally associated with stronger pro-environmental attitudes and food waste reduction behaviours [37, 38], although some studies have found that highly educated households may also be less sustainable [39]. Age has been more consistently associated with food waste outcomes, with older consumers tending to waste less food [40]. These conventional factors, like gender, education, and age of respondents and household food waste behaviour are also incorporated.

Based on these theoretical foundations and empirical findings, the conceptual framework of this study incorporates a set of psychological, behavioural, and demographic constructs previously identified in the literature as potential drivers of household food waste. These hypothesised relationships are visualised in Fig. 1 and tested below.

Data and empirical method

Sample design

The household survey employed a multi-stage sampling strategy to ensure broad spatial coverage and representativeness across urban Ho Chi Minh City. In the first stage, residential clusters were defined based on the population distribution across city areas, resulting in the selection of 50 residential clusters spanning 25 wards. Updated household lists, including addresses of household heads, were obtained from local administrative offices. In the second stage, households within each cluster were randomly selected, with approximately 20 households being interviewed per cluster, based on their willingness to participate.

This procedure produced an initial target sample of approximately 1,000 households (50 clusters × 20 households), forming the empirical basis for the analysis. Sampling weights were not applied, as the primary objective of the study was to model behavioural relationships rather than to produce population-weighted estimates. In line with standard practice in structural equation modelling, unweighted estimation is appropriate when the focus is on testing theoretical relationships among constructs rather than estimating population parameters [41, 42]. To account for potential heterogeneity across households, key demographic variables were included as controls in the models. The survey design

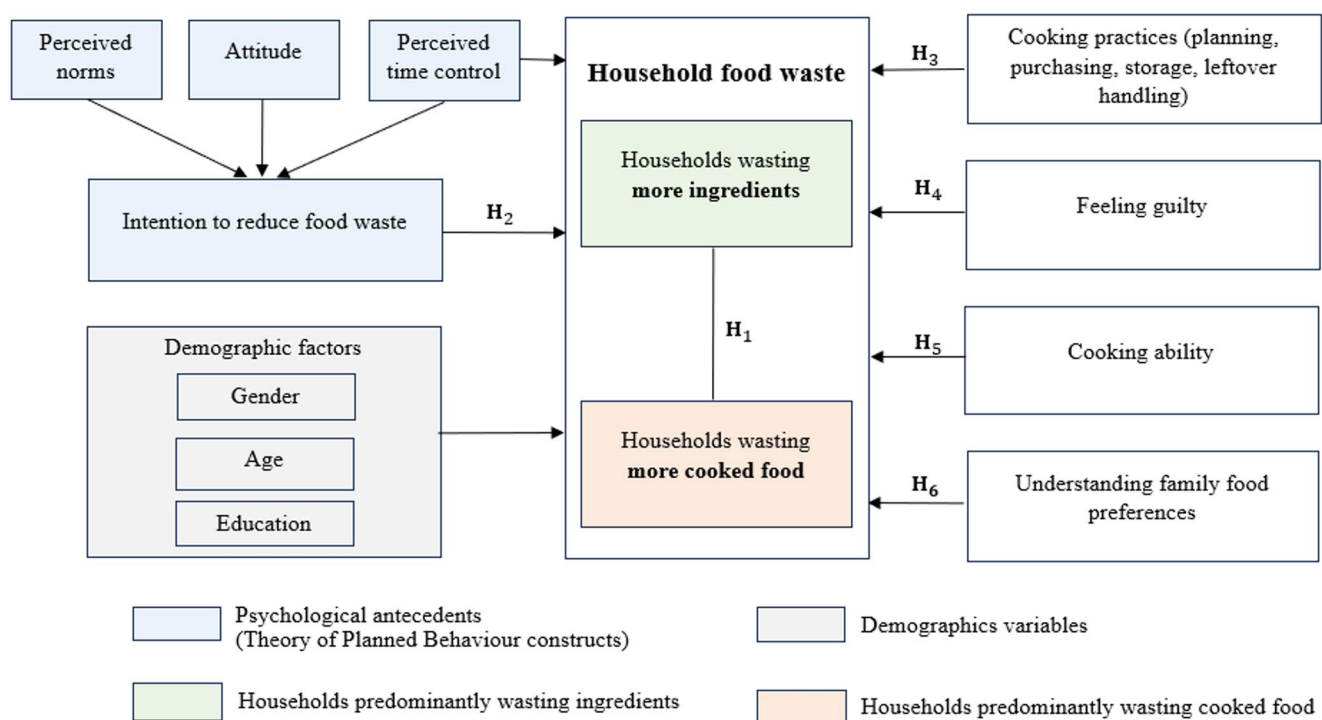


Fig. 1 Conceptual framework for modelling urban household food waste behaviours using CB-SEM, focusing on behavioural drivers among households that predominantly waste ingredients versus those that waste cooked food

and data collection process have been described in detail in a related publication using the same dataset [39].

Data collection

Face-to-face interviews were conducted with households in Ho Chi Minh City between September and October 2023 across the selected residential clusters. Enumerators received standardised training before fieldwork, including detailed guidance on neutral question delivery, avoiding leading language, and reassuring respondents that there were no “right” or “wrong” answers -- an approach widely

recommended to reduce social desirability and recall bias in survey-based research [43, 44]. To reduce recall and social desirability bias, most behavioural measures were captured using Likert-scale statements (1–7) that focused on typical practices rather than exceptional behaviours, and interviews were conducted in a non-judgemental manner.

Respondents were informed that their answers were anonymous and used solely for research purposes, which helped encourage honest reporting. The spatial distribution of participating households across districts is shown in Fig. 2, illustrating broad geographic coverage of the survey sample within urban Ho Chi Minh City. Households that did

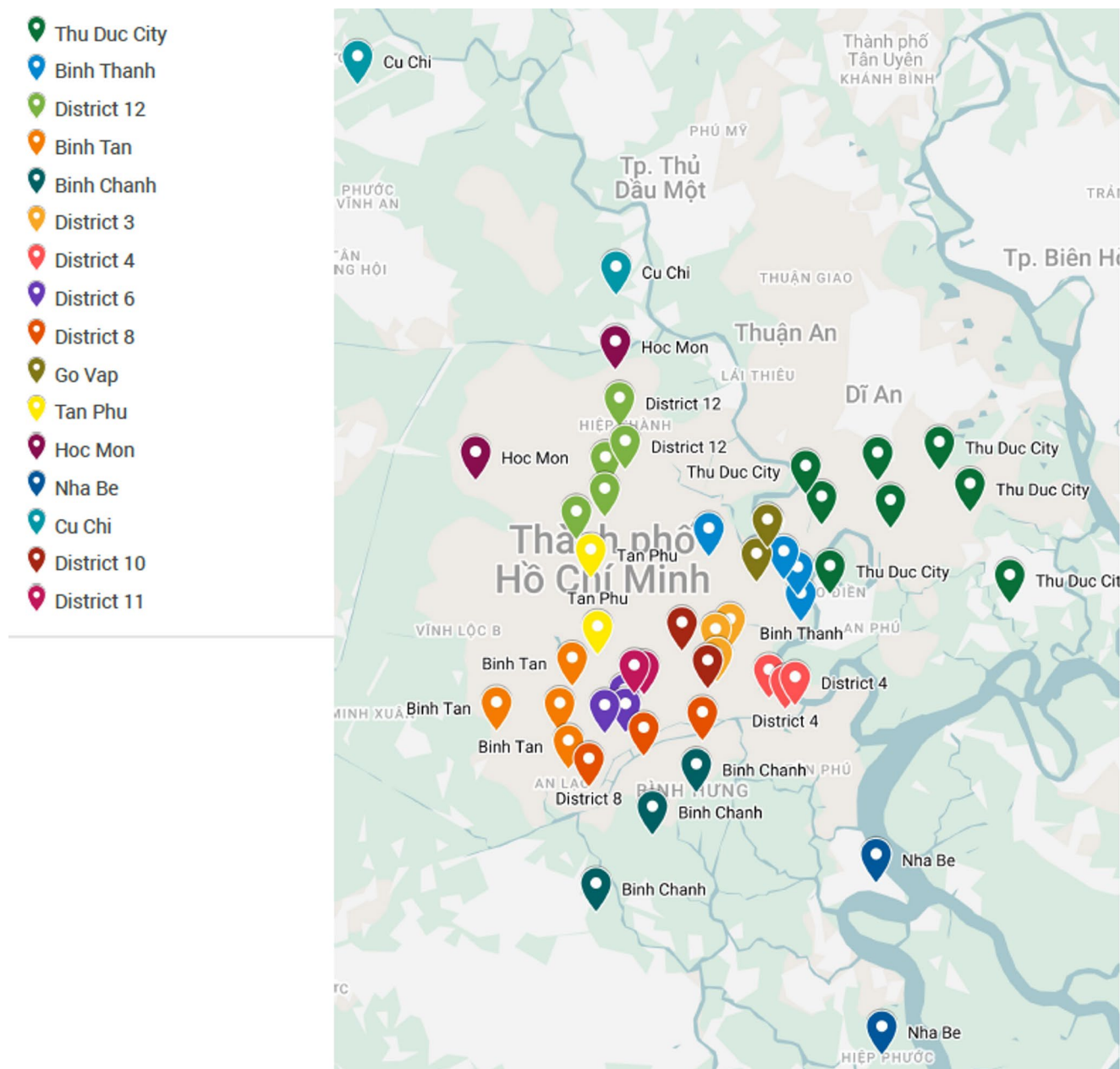


Fig. 2 Spatial distribution of surveyed households across Ho Chi Minh City [39]

not answer questions regarding household food waste levels (the primary endogenous variable in the SEM analysis) were excluded from further analysis.

Data were collected using smartphones and tablets via the CommCare platform, with built-in validation rules and skip logic programmed during questionnaire design to minimise missing data. Little's MCAR Test was conducted to evaluate the null hypothesis that "*missing data is Missing Completely at Random (MCAR)*". The results (Chi-square=0.000, DF=203) indicate that we failed to reject the null hypothesis, suggesting it is reasonable to treat the missing data as missing at random. After data cleaning and exclusion of incomplete cases, a final analytical sample of 642 households was retained.

Methodology

We employed structural equation modelling (SEM) techniques to analyse the behavioural drivers of household food waste during home cooking. SEM enables the simultaneous analysis of multiple dependent and independent variables, incorporates latent constructs through measurement models, and allows for the assessment of mediation and moderation effects [42]. Two main SEM approaches were used in this study: Covariance-Based SEM (CB-SEM) and Variance-Based SEM (VB-SEM), with the choice determined by research objectives, sample size, and the level of model complexity [45]. CB-SEM is typically used for confirmatory analysis and theory testing with large sample sizes, while VB-SEM is preferred for predictive modelling and exploratory analysis with small sample sizes [45]. CB-SEM is typically used for theory testing and confirmatory analysis in large samples, while VB-SEM is preferred for exploratory modelling and prediction when working with small or non-normal samples.

CB-SEM was performed in AMOS 29 to test theory-driven hypotheses within two food-wasting household segments: households predominantly wasting **raw ingredients** and those wasting more **cooked food**. Given the confirmatory nature of this analysis and the relatively large sample size, CB-SEM was deemed appropriate. In addition, an exploratory VB-SEM model was conducted using Smart-PLS 4 to analyse a smaller subset ($n=113$) of **zero-waste households** (those reporting no food waste during home cooking).

These households were identified based on self-reported zero food waste following a structured validation process embedded in the survey administration. Respondents were first asked to report the average percentage of purchased food (meat, fruits, and vegetables) discarded by the household. When a respondent reported 0% food waste, subsequent

questions on the composition of food waste were treated as not applicable but were used by enumerators as clarification prompts to confirm the internal consistency of the response. Specifically, respondents were asked to reflect on whether this zero-waste report implied that they never discarded any purchased meat/fruits/vegetables and whether food was consistently managed to avoid both ingredient waste during cooking and cooked food waste after meals. Respondents were allowed to revise their initial estimate if needed. Only households that maintained a zero-waste response after this clarification process were classified as zero-waste households.

This subgroup was not anticipated at the questionnaire design stage, as the primary analytical focus was initially on comparing ingredient-wasting and cooked-food-wasting households using CB-SEM. However, the emergence of a non-trivial group of self-reported zero-waste households during data collection motivated an additional exploratory VB-SEM analysis to investigate potential behavioural patterns associated with waste prevention. VB-SEM was selected due to its suitability for small samples and exploratory modelling. We acknowledge that the identification of zero-waste households is based on self-reported data and may therefore be subject to underreporting or social desirability bias. Accordingly, findings from the VB-SEM analysis are interpreted as exploratory, offering indicative insights into waste-prevention behaviours rather than definitive causal conclusions. Together, this dual-method approach enabled a comprehensive investigation of household food waste behaviours across three distinct household types (ingredient-wasting households, cooked-food-wasting households, and zero-waste households) while appropriately distinguishing between confirmatory and exploratory analytical objectives.

Results

The empirical analysis begins with Covariance-Based Structural Equation Modelling (CB-SEM) to test key behavioural hypotheses, including model diagnostics, measurement validation, structural estimation, and group-level comparisons. This is followed by an exploratory Variance-Based SEM (VB-SEM) to investigate food waste prevention behaviours among a distinct segment of zero-waste urban households—those who reported not wasting any food during home cooking. Refer to Table 9, Appendix A, for the descriptive statistics and measurement details of all constructs.

Modelling behavioural drivers of urban household food waste using CB-SEM

CB-SEM typically assumes multivariate normality for its maximum likelihood estimation algorithm [46]. However, the test results revealed that multivariate normality was not met in this study's database: multivariate kurtosis = 333.39, much larger than 5. In such circumstances, bias-corrected bootstrapping is widely recommended as a robust estimation approach for CB-SEM, as it relaxes distributional assumptions and provides more reliable standard errors and confidence intervals under non-normal conditions [47, 48]. Accordingly, we employed bias-corrected bootstrapping in this study to ensure robust inference despite violations of multivariate normality.

Confirmatory Factor Analysis was conducted to evaluate the measurement model. Observed indicators with factor loadings below 0.50 were removed during validation. The revised measurement model was then tested for adequacy. As shown in Table 1, the chi-square (χ^2) test yielded a *p*-value less than 0.05, indicating a statistically significant difference between the hypothesised and observed models. However, this is a common issue in Covariance-Based SEM (CB-SEM), especially with large sample sizes [42]. In such cases, a significant chi-square does not necessarily imply model misspecification. Therefore, additional fit indices were considered. The model demonstrates acceptable test statistics, including $\chi^2/DF = 2.757$ (below the recommended threshold of 3); and all GFI, CFI, IFI, and TLI values exceeding the recommended threshold of 0.90; and RMSEA = 0.052.

Assessing the reliability and validity of latent constructs is crucial for ensuring that the data used in SEM accurately represents the underlying constructs, which is essential for producing meaningful and trustworthy results [42]. Upon evaluating the reliability of the constructs, we find that the Cronbach's Alpha values for nearly all latent constructs exceed 0.8, with only two constructs being less than 0.8 (see Table 10, Appendix A). Additionally, all composite reliability (CR) values for the latent variables are above (see Table 2), which minimises possible concerns over construct reliability.

All factor loadings are statistically significant regarding convergent validity, with standardised loadings exceeding 0.5 (with most above 0.7). Furthermore, the CR values are greater than the average variance extracted (AVE) values, and all AVE values are above 0.5 (see Table 2). These results suggest no concerns with the constructs' convergent

validity. Table 2 shows that, for nearly all constructs, the maximum shared squared variance (MSV) values are lower than the average variance extracted (AVE) values.

Additionally, the square roots of the AVE values are generally greater than 0.7 and exceed the absolute values of the correlations with other factors, indicating acceptable discriminant validity for most latent constructs. Two exceptions were observed for Attitudes and Perceived Norms, where MSV values exceeded AVE and the square roots of AVE were lower than their mutual correlation, suggesting potential overlap between these constructs.

To further assess discriminant validity, the heterotrait–monotrait ratio (HTMT) was examined. HTMT values for most construct pairs were below the recommended threshold of 0.90 (see Table 11, Appendix A), indicating satisfactory discriminant validity. The HTMT value for the Attitudes–Perceived Norms pair was 0.909, marginally exceeding the threshold and suggesting a borderline discriminant validity concern. Although attitudes and perceived norms exhibit a potential discriminant validity concern at the construct level, item-level diagnostics indicate that observed indicators load strongly on their intended constructs, with no substantive cross-loadings (see Fig. 5, Appendix A). This indicates that the constructs remain empirically distinct.

To further examine whether this overlap reflected a lack of empirical distinction, we estimated an alternative model in which Attitudes and Perceived Norms were combined into a single latent construct as a robustness check. This alternative specification showed poorer overall model performance and reduced explanatory power relative to the original model and was therefore not retained. Given the confirmatory focus of the analysis, detailed results of this alternative specification are not reported to preserve model parsimony.

Moreover, consistent with the Theory of Planned Behaviour, Attitudes and Perceived Norms represent conceptually distinct psychological processes—individual evaluations versus perceived social pressure—and are therefore retained as separate constructs. Taken together, these results suggest that while Attitudes and Perceived Norms are closely related in this empirical context, they remain theoretically and empirically distinguishable.

Results of the general household food waste model

CB-SEM demonstrated an acceptable-to-good fit based on multiple fit indices (see Table 3). The standardised root mean square residual (SRMR) was 0.0348, well below the

Table 1 Model fit assessment values of the measurement model

Indices	<i>p</i> -value of χ^2	χ^2	DF	χ^2/DF	GFI	CFI	IFI	TLI	RMSEA
Recommended threshold	>0.05	-	-	<3	>0.9	>0.9	>0.9	>0.9	<0.08
Measurement model	0.000	879.4	319	2.757	0.919	0.955	0.955	0.9	0.052

Table 2 Checking the reliability and validity of the latent constructs

	CR	AVE	MSV	intention	attitudes	per- ceived norms	per- ceived time control	feeling guilty	storage practices	planning behaviours	leftover handling behaviours	cooking ability	under- standing family food preferences	house- hold food waste
Intention	0.934	0.825	0.402	0.908										
Attitudes	0.896	0.742	0.815	-0.526	0.861									
Perceived norms	0.760	0.616	0.815	0.602	-0.903	0.785								
Perceived time control	0.891	0.805	0.032	0.100	0.120	-0.157	0.897							
Feeling guilty	0.917	0.787	0.531	0.634	-0.630	0.729	-0.054	0.887						
Storage practices	0.819	0.605	0.220	0.315	-0.423	0.469	-0.157	0.417	0.778					
Planning behaviours	0.937	0.881	0.168	0.154	-0.166	0.135	-0.016	0.173	0.410	0.939				
Leftover handling behaviours	0.817	0.692	0.169	0.098	-0.165	0.210	0.076	0.266	0.411	0.099	0.832			
Cooking ability	0.925	0.860	0.554	-0.018	0.253	-0.315	0.139	-0.130	-0.255	-0.156	-0.065	0.928		
Understanding family food preferences	0.726	0.571	0.554	0.280	-0.520	0.598	-0.179	0.402	0.450	0.273	0.201	-0.744	0.755	
Household food waste	0.828	0.707	0.471	-0.421	0.686	-0.668	0.170	-0.469	-0.403	-0.210	-0.229	0.266	-0.537	0.841

Underlined values indicate cases where MSV exceeds AVE, suggesting potential issues with discriminant validity.

recommended threshold of 0.08 [49]. The chi-square ratio $\chi^2/DF = 2.938$, meeting the criterion of < 3 . Additionally, all incremental fit indices (GFI, CFI, IFI, and TLI) exceeded 0.90, and the root mean square error of approximation (RMSEA) was 0.055, remaining within the acceptable range (< 0.08).

The model explains 71% of the variance in household food waste ($R^2 = 0.71$), manifesting strong explanatory power. The SEM results (see Table 4) indicate that attitudes ($\beta = 1.992$, $p = 0.001$) and perceived norms ($\beta = 2.626$, $p = 0.000$) significantly predict intention to reduce food waste. However, the intention ($\beta = -0.008$, $p = 0.980$) does not significantly influence household food waste. Among the significant factors, understanding family food preferences ($\beta = -1.831$, $p = 0.001$) and cooking ability ($\beta = -1.244$, $p = 0.001$) were the strongest, both negatively associated with household food waste. This suggests that in urban household contexts, better knowledge of family food preferences and stronger cooking ability are critical skills for minimising food waste. Additionally, waste cost concerns ($\beta = -0.180$, $p = 0.056$) showed a marginally negative association with household food waste, suggesting that households are more mindful of food costs and tend to generate less waste. Interestingly, traditional food management practices such as storage, planning, and leftover handling showed no significant association with overall food waste, possibly reflecting the influence of daily shopping habits in urban areas. Likewise, demographic characteristics — age, education, and gender — did not emerge as significant predictors, reinforcing the behavioural rather than socio-demographic nature of food waste patterns in this urban sample.

Given evidence of multivariate non-normality in the dataset, statistical inference was based on resampling-based procedures. In this study, bias-corrected (BC) bootstrapping was employed as the primary approach, consistent with established practice in applied CB-SEM under non-normal data conditions [47, 48]. To further assess the stability of the estimated parameters, percentile bootstrapping (PC) was implemented as a complementary robustness check, following methodological guidance indicating its favourable performance when resampling-based methods are feasible [50].

The model was re-estimated using percentile bootstrap confidence intervals with 5,000 resamples (see Table 12, Appendix A). The results obtained using the PC method were substantively identical to those from the BC bootstrap, with consistent coefficient directions, magnitudes, and levels of statistical significance across all structural paths. This convergence provides additional reassurance regarding the robustness of the reported CB-SEM findings.

Table 3 Model fit assessment values of the General Household Food Waste Model

Indices	p -value of χ^2	χ^2	DF	χ^2/DF	GFI	CFI	IFI	TLI	RMSEA
Recommended threshold	>0.05	-	-	<3	>0.9	>0.9	>0.9	>0.9	<0.08
Measurement model	0.000	975.4	332	2.938	0.911	0.948	0.949	0.927	0.055

Table 4 Standardised Regression Weights of the General Household Food Waste Model

Paths	General Household Food Waste Model ($R^2 = 0.71$)			
	Estimate (β)	95% CI Lower	95% CI Upper	p -value
Intention $\leftarrow$ Attitudes	1.992***	0.465	15.240	0.001
Intention $\leftarrow$ Perceived Time Controls	0.296***	0.175	1.020	0.002
Intention $\leftarrow$ Perceived Norms	2.626***	1.115	15.835	0.000
Household Food Waste $\leftarrow$ Intention	-0.008	-0.212	0.393	0.980
Household Food Waste $\leftarrow$ Perceived Time Controls	0.085	-0.085	0.285	0.288
Household Food Waste $\leftarrow$ Feeling Guilty	0.179	-0.108	0.590	0.225
Household Food Waste $\leftarrow$ Storage Practices	0.123	-0.151	0.507	0.352
Household Food Waste $\leftarrow$ Planning Behaviours	0.099	-0.086	0.389	0.340
Household Food Waste $\leftarrow$ Leftover Handlings	0.028	-0.171	0.316	0.786
Household Food Waste $\leftarrow$ Cooking Ability	-1.244***	-2.447	-0.590	0.001
Household Food Waste $\leftarrow$ Understanding Family Food Preferences	-1.831***	-3.202	-1.039	0.001
Household Food Waste $\leftarrow$ Waste Cost Concerns	-0.180*	-0.509	-0.004	0.056
Household Food Waste $\leftarrow$ Buying Big Packages	0.211*	-0.016	0.609	0.071
Household Food Waste $\leftarrow$ Female	0.066	-0.090	0.291	0.421
Household Food Waste $\leftarrow$ Education	0.112	-0.041	0.293	0.144
Household Food Waste $\leftarrow$ Age	-0.050	-0.235	0.123	0.556

*** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$

Results from the multi-group CB-SEM analysis by dominant waste type

Among the 642 surveyed households, 419 provided valid responses to the question distinguishing whether they primarily waste raw ingredients or cooked food. Respondents were asked to estimate the percentage share of total household food waste attributable to raw ingredients versus cooked food, assuming total food waste equalled 100%. Based on these self-reported proportions, households were categorised into two groups: Group 1 ($n=246$), comprising households reporting a higher proportion of ingredient waste, and Group 2 ($n=173$), comprising households reporting a higher proportion of cooked food waste. To assess whether these two groups differed meaningfully in waste composition, we examined the distribution of ingredient and cooked food waste shares across groups. Ingredient-wasting households reported a mean of 83.7% ingredient waste (median=90%), with cooked food accounting for only 16.3% on average, indicating a strongly skewed waste profile. Conversely, cooked-food-wasting households reported a mean of 81.4% cooked food waste (median=80%), with substantially lower shares of ingredient waste. These distributions confirm that the two groups represent distinct dominant waste patterns rather than balanced or overlapping profiles (see Table 13, Appendix A). A moderation analysis

was conducted using this “group” variable within the SEM framework to assess whether structural paths differed across these segments. While running the model with this moderator, we applied constraints, setting the structural weights of Group 1 equal to Group 2, and tested the null hypothesis: “Assuming the unconstrained model to be correct.” The test results (p -value=0.000) failed to reject the null hypothesis, confirming that the unconstrained model is valid. This result supports the use of separate models to account for segment-specific behavioural pathways in household food waste.

Both group-specific SEM models showed acceptable-to-good model fit (CMIN/DF=1.851, IFI=0.933, GFI=0.852, TLI=0.904, CFI=0.931, RMSEA=0.045). The model for households with higher ingredient waste explained 59% of the variance ($R^2 = 0.59$), whereas the model for households with higher cooked food waste accounted for 75% of the variance ($R^2 = 0.75$). This indicates stronger predictive power for households that waste more cooked food. Table 5 reports the SEM results. It is seen that, consistent with the baseline model, attitudes and perceived norms significantly predicted intention. However, the intention remains a non-significant predictor of household food waste in both moderated models. Similarly, understanding family food preferences and cooking ability are the strongest predictors, both negatively associated with household food waste. In contrast, storage practices, planning behaviours, and

Table 5 Standardised Regression Weights of the Moderated Household Food Waste Models

Paths	Households wasting more ingredients ($R^2 = 0.59$)				Households wasting more cooked food ($R^2 = 0.75$)			
	Estimate (β)	95% CI Lower	95% CI Upper	<i>p</i> -value	Estimate (β)	95% CI Lower	95% CI Upper	<i>p</i> -value
Intention $\leftarrow$ Attitudes	15.361***	8.132	...	0.000	0.317**	0.011	1.381	0.042
Intention $\leftarrow$ Perceived Time Controls	−0.131	...	0.344	0.382	0.085*	−0.016	0.248	0.091
Intention $\leftarrow$ Perceived Norms	15.677***	8.481	...	0.000	1.129***	0.823	2.047	0.003
Household Food Waste $\leftarrow$ Intention	0.107	−0.220	0.740	0.673	−0.278	−0.991	0.211	0.183
Household Food Waste $\leftarrow$ Perceive Time Controls	0.100	−0.390	0.289	0.529	0.106	−0.242	2.256	0.333
Household Food Waste $\leftarrow$ Feeling Guilty	0.052	−0.239	0.420	0.627	0.988***	0.332	...	0.008
Household Food Waste $\leftarrow$ Storage Practices	0.028	−0.295	0.490	0.987	1.126	−0.114	8.124	0.801
Household Food Waste $\leftarrow$ Planning Behaviours	0.014	−0.232	0.486	0.953	−0.155	−2.967	0.697	0.418
Household Food Waste $\leftarrow$ Leftover Handlings	0.040	−0.222	0.408	0.966	−0.747	−8.764	0.142	0.111
Household Food Waste $\leftarrow$ Cooking Ability	−0.617**	−2.019	−0.066	0.013	−2.119**	−12.209	−0.217	0.013
Household Food Waste $\leftarrow$ Understanding Family Food Preferences	−1.138**	−3.041	−0.392	0.013	−3.119***	−14.237	−0.681	0.001
Household Food Waste $\leftarrow$ Waste Cost Concerns	−0.158	−0.636	0.037	0.129	−0.127	−1.589	2.182	0.742
Household Food Waste $\leftarrow$ Buying Big Packages	0.174	−0.090	0.952	0.239	0.874**	0.046	4.864	0.031
Household Food Waste $\leftarrow$ Female	−0.058	−0.276	0.179	0.544	0.058	−0.523	1.431	0.718
Household Food Waste $\leftarrow$ Education	0.091	−0.234	0.303	0.380	0.334*	−0.066	2.512	0.098
Household Food Waste $\leftarrow$ Age	−0.207**	−0.437	−0.015	0.047	0.363*	−0.037	...	0.092

*** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$

leftover handling continue to show no significant impact in either moderated model. In the ingredient-wasting model, age becomes a significant predictor ($\beta = -0.207$, $p = 0.047$), suggesting older individuals waste less. However, in the cooked-food-wasting model, age loses significance, while guilt ($\beta = 0.988$, $p = 0.008$) emerges as a significant positive predictor of food waste. Additionally, buying big packages ($\beta = 0.874$, $p = 0.031$) significantly increases food waste in these households.

Exploratory modelling of zero-waste urban households using VB-SEM

The subgroup of 113 households that reported zero food waste offers a unique opportunity to explore the behavioural patterns associated with sustained waste prevention in home cooking—an important contribution to sustainable consumption in rapidly urbanising contexts. To capture this, we conducted an exploratory VB-SEM without specifying a single dependent variable. Instead, we examined inter-relationships among key psychological, behavioural, and food management constructs to gain insight into the internal dynamics of these high-performing households. This model complements the main SEM analysis and provides initial insights, offering preliminary guidance for future research on best-practice behaviours in sustainable home cooking, with broader implications for urban food governance and sustainable city planning.

Conceptual path structure for the exploratory VB-SEM model

Building on the literature and the hypothesis-testing results of the CB-SEM models, we proposed a three-layer conceptual structure for the VB-SEM analysis Fig. 3. This includes: (1) Drivers, (2) Enablers, and (3) Behaviours.

Layer 1 – Drivers: Motivational and Sociodemographic Influences.

This layer captures the foundational motivational and contextual factors that shape intentions and behaviours related to food waste reduction. It includes perceived norms, attitude, and perceived time control, representing key psychological motivators as well as gender, age, and education as important sociodemographic influences.

Layer 2 – Enablers: Intentions and Capabilities to Act.

The second layer consists of factors that facilitate the translation of intention into action. It includes the intention to reduce food waste, along with two key constructs—cooking ability and understanding of family food preferences—which were identified as significant predictors of household food practices in the hypothesis-testing results of the CB-SEM models presented earlier. Together, these enablers represent the knowledge, skills, and motivation required to effectively manage food in the household.

Layer 3 – Behaviours: Psychological and Practical Food Waste Actions.

The final layer comprises both psychological responses and observable food management behaviours: feeling guilty about wasting food, leftover handling, planning behaviour,

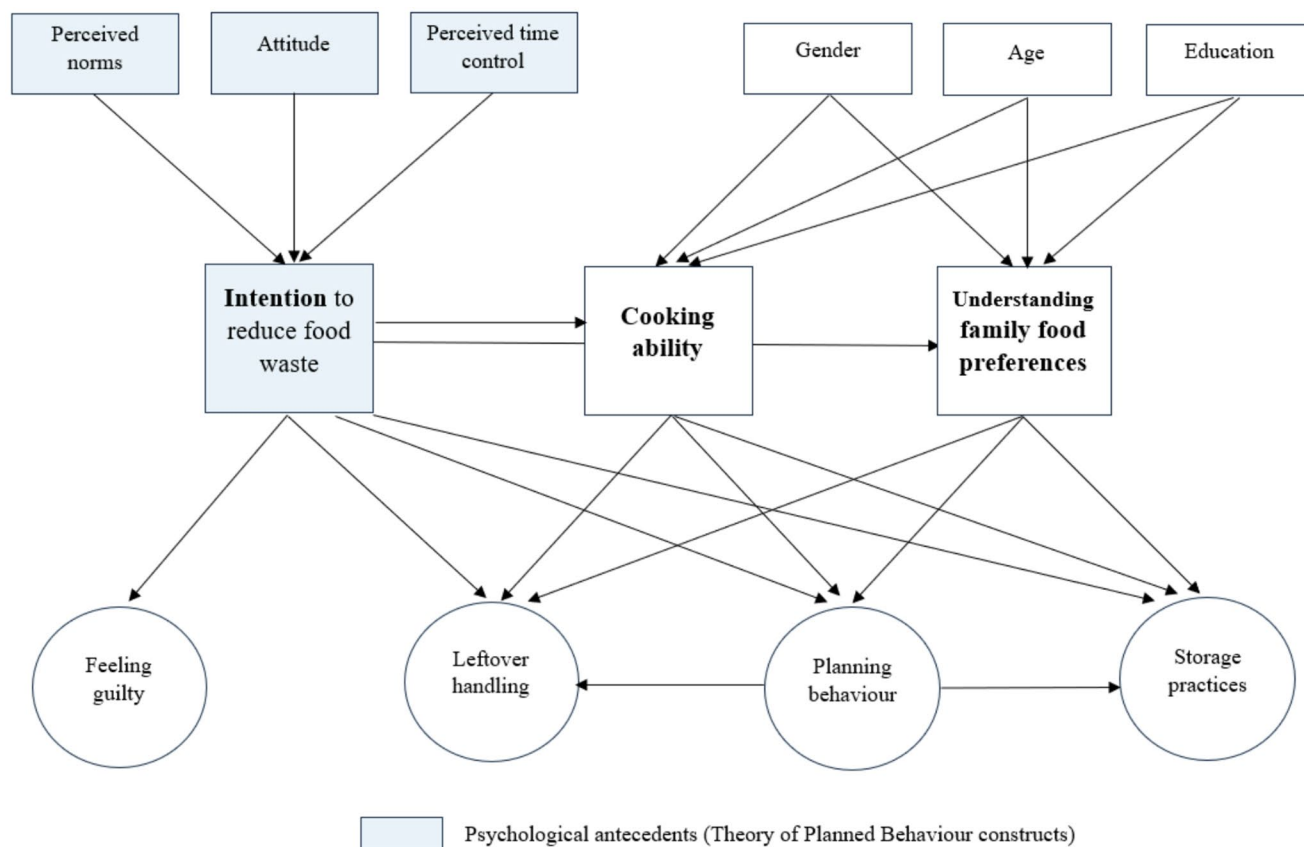


Fig. 3 Conceptual three-layer framework (drivers–enablers–behaviours) for modelling zero-waste household behaviours using VB-SEM

and storage practices. These practices reflect the concrete strategies and internal reactions that households adopt to minimise food waste, encompassing both cognitive-emotional regulation and day-to-day management routines.

Measurement model assessment for VB-SEM

Before conducting the VB-SEM, the reliability and validity of the measurement model were assessed by established guidelines. Constructs were evaluated using Cronbach's alpha, composite reliability (CR), and average variance extracted (AVE), with the following thresholds: Cronbach's alpha ≥ 0.70 , composite reliability ≥ 0.70 , and AVE ≥ 0.50 [45]. In the initial assessment (see Table 14, Appendix A), most constructs met these criteria. However, the “leftover handling” construct showed poor internal consistency (Cronbach's alpha = 0.369, AVE = 0.464) and was therefore excluded from the model. Additionally, the “cooking ability” construct included a low-loading item (cooking_ability_3, loading = 0.13), which was removed to improve construct reliability. After these adjustments, all remaining constructs demonstrated satisfactory reliability and convergent validity (see Table 6).

Table 6 Construct validity and reliability (VB-SEM)

Constructs	Cronbach's Alpha (threshold: ≥ 0.70)	CR (threshold: ≥ 0.70)	AVE (threshold: ≥ 0.70)
Attitude	0.942	0.963	0.897
Cooking ability	0.926	0.964	0.931
Feeling guilty	0.963	0.976	0.931
Intention to reduce food waste	0.962	0.975	0.929
Perceived norms	0.848	0.909	0.772
Perceived time controls	0.936	0.968	0.939
Planning behaviours	0.953	0.977	0.955
Storage practices	0.885	0.922	0.749
Understanding family food preferences	0.869	0.937	0.882

Discriminant validity was assessed HTMT, applying a threshold of <0.90 . The results (see Table 15, Appendix A) showed that all construct pairs met this criterion, except for the relationship between intention and feeling guilty (HTMT=0.930). Given the conceptual proximity of these constructs—where feelings of guilt may reasonably inform behavioural intentions—this overlap was theoretically justified, and both constructs were retained in the final model.

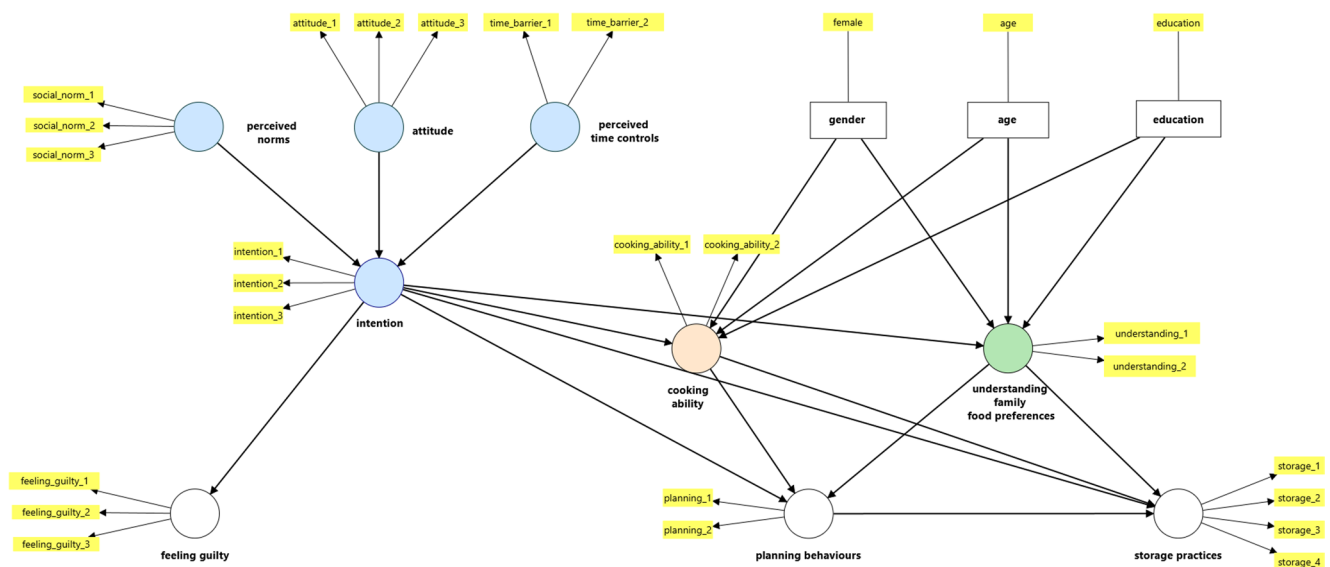


Fig. 4 VB-SEM path analysis for the zero-waste household group

Table 7 Results of VB-SEM path analysis for the zero-waste household group

Paths	Coefficient (β)	2.5%	97.5%	T statistics	P values
Age $\rightarrow$ Cooking Ability	-0.188**	-0.364	-0.015	2.125	0.034
Age $\rightarrow$ Understanding Family Food Preferences	0.153*	-0.014	0.327	1.723	0.085
Attitude $\rightarrow$ Intention	0.263**	0.083	0.587	2.071	0.038
Cooking Ability $\rightarrow$ Planning Behaviours	0.091	-0.075	0.251	1.106	0.269
Cooking Ability $\rightarrow$ Storage Practices	-0.088	-0.256	0.064	1.073	0.283
Education $\rightarrow$ Cooking Ability	-0.049	-0.255	0.154	0.472	0.637
Education $\rightarrow$ Understanding Family Food Preferences	0.221***	0.065	0.369	2.863	0.004
Gender $\rightarrow$ Cooking Ability	-0.351*	-0.753	0.025	1.781	0.075
Gender $\rightarrow$ Understanding Family Food Preferences	0.520**	0.096	0.920	2.436	0.015
Intention $\rightarrow$ Cooking Ability	-0.288***	-0.453	-0.110	3.237	0.001
Intention $\rightarrow$ Feeling Guilty	0.895***	0.835	0.939	33.614	0.000
Intention $\rightarrow$ Planning Behaviours	0.234***	0.067	0.402	2.722	0.007
Intention $\rightarrow$ Storage Practices	0.506***	0.301	0.704	4.898	0.000
Intention $\rightarrow$ Understanding Family Food Preferences	0.450***	0.301	0.581	6.232	0.000
Perceived Norms $\rightarrow$ Intention	0.608***	0.291	0.784	4.849	0.000
Perceived Time Controls $\rightarrow$ Intention	0.015	-0.062	0.098	0.371	0.711
Planning Behaviours $\rightarrow$ Storage Practices	0.308***	0.089	0.504	2.923	0.003
Understanding Family Food Preferences $\rightarrow$ Planning Behaviours	0.338***	0.147	0.528	3.488	0.000
Understanding Family Food Preferences $\rightarrow$ Storage Practices	0.005	-0.192	0.222	0.045	0.964

*** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$

Results from the VB-SEM model

After removing the “leftover handling” construct and adjusting the “cooking ability” construct, the VB-SEM model was estimated for the zero-waste household group ($n = 113$). The final model is presented in Fig. 4, and the estimated path coefficients, bootstrapped t-values, and p-values (5,000 subsamples) are reported in Table 7.

The VB-SEM results indicate that behavioural intention plays a central role in the zero-waste household group. Intention significantly predicts multiple downstream constructs,

including planning behaviours ($\beta = 0.234$, $p = 0.007$), storage practices ($\beta = 0.506$, $p < 0.001$), understanding family food preferences ($\beta = 0.450$, $p < 0.001$), and feeling guilty ($\beta = 0.895$, $p < 0.001$). Psychological antecedents also show significant associations with intention: attitude ($\beta = 0.263$, $p = 0.038$) and perceived norms ($\beta = 0.608$, $p < 0.001$) are positively related to intention, whereas perceived time controls do not exhibit a significant effect ($\beta = 0.015$, $p = 0.711$). Within the behavioural layer, planning behaviours are positively associated with storage practices ($\beta = 0.308$, $p = 0.003$). Understanding family food preferences is positively related

to planning behaviours ($\beta=0.338, p<0.001$), but not to storage practices directly ($\beta=0.005, p=0.964$). Cooking ability does not show significant direct associations with planning behaviours ($\beta=0.091, p=0.269$) or storage practices ($\beta=-0.088, p=0.283$).

Taken together, these results describe the internal behavioural structure of zero-waste households and provide a basis for comparison with the CB-SEM findings for food-wasting household groups.

Discussions

This study applied Covariance-Based Structural Equation Modelling (CB-SEM) to examine the behavioural drivers of household food waste during home cooking, with hypothesis testing results summarised in Table 8. The findings demonstrate that structural relationships differ significantly between households that mainly waste raw ingredients and those that mainly waste cooked food. The rejection of the constrained multi-group model ($p=0.000$) confirms H_1 , that a single, uniform behavioural structure is insufficient for explaining household food waste behaviours. This result aligns with emerging literature that calls for more differentiated, pathway-specific approaches to food waste analysis rather than one-size-fits-all models.

Across all CB-SEM models, intention to reduce food waste did not significantly predict actual household food waste, confirming H_2 was not supported. This finding reinforces the well-documented intention–behaviour gap in food waste research, where positive attitudes and strong intentions do not necessarily translate into reduced waste outcomes. While many previous studies focus primarily on intention as a proxy for sustainable behaviour, the present results highlight the limitations of intention-centric approaches and underscore the importance of examining concrete practices and household capabilities when designing food waste interventions.

Contrary to expectations and findings from other contexts, most home cooking practices—specifically planning behaviours, storage practices, and leftover handling—did not exhibit significant associations with household food waste, leading to limited support for H_3 . A plausible explanation lies in the urban food environment of Ho Chi Minh City, where households commonly purchase fresh ingredients almost daily due to the high density of neighbourhood food retailers. Frequent shopping reduces reliance on long-term storage, planning, and systematic leftover management, thereby diminishing the relevance of these practices as waste-reduction mechanisms. This contextual explanation highlights the importance of place-specific food system characteristics and cautions against directly transferring

Table 8 Hypothesis testing results from the CB-SEM models

Hypothesis		General model	Households wasting more ingredients	Households wasting more cooked food
H_1 : The structural paths linking constructs to food waste outcomes differ significantly between households that waste more raw ingredients and those that waste more cooked food.		N/A	Significant	
H_2 : The greater the intention to reduce food waste, the smaller the amount of household food waste.		Not significant	Not significant	Not significant
H_3 : Home cooking practices—including food planning, purchasing, storage, and leftover handling—are significantly associated with household food waste.	Better planning - lower food waste.	Not significant	Not significant	Not significant
	Purchasing large packages - higher food waste.	Not significant	Not significant	Significant
	Better storage practices - lower food waste.	Not significant	Not significant	Not significant
	Better leftover handling practices - lower food waste.	Not significant	Not significant	Not significant
H_4 : Greater guilt is associated with lower food waste.		Not significant	Not significant	Significant (opposite sign)
H_5 : Greater cooking skills are linked to reduced household food waste.		Significant	Significant	Significant
H_6 : A deeper understanding of family needs is linked to reduced household food waste.		Significant	Significant	Significant

testing outcomes at the 5% significance level.

behavioural interventions across settings without considering local food acquisition routines.

In contrast, capability-based factors emerged as the most robust and consistent predictors of household food waste. Both cooking ability (H_5) and understanding family food preferences (H_6) were significantly and negatively associated with food waste across the general model and both waste-type subgroups. These findings reinforce prior research emphasising the central role of practical skills and household coordination in food waste prevention. Rather than focusing solely on awareness or motivation, the results suggest that strengthening everyday cooking competence and improving alignment between meal preparation and household needs are critical levers for reducing waste in urban households.

The analysis also reveals meaningful differences between ingredient-wasting and cooked-food-wasting households. Purchasing large package sizes was significantly associated with higher food waste only in the cooked-food-wasting group, partially supporting H_3 . This suggests that bulk purchasing may not necessarily lead to ingredient spoilage but can encourage over-preparation and excessive cooking, particularly in urban households responding to timesaving or cost-saving incentives. Since cooked food waste embodies additional embedded resources—such as energy, water, and labour—this pathway represents a particularly unsustainable form of waste and warrants targeted policy attention.

Emotional drivers further differentiate waste pathways. While guilt did not significantly reduce food waste in the general or ingredient-wasting models, it was positively associated with food waste in the cooked-food-wasting group, contradicting H_4 . This counterintuitive finding challenges the assumption that guilt functions solely as a constructive motivator for pro-environmental behaviour. Post-rationalisation theory offers a potential explanation, suggesting that individuals may experience guilt after wasting food but rationalise the behaviour without altering future practices. This finding underscores the risks of relying on guilt-based messaging without providing practical means for behaviour change.

Taken together, the CB-SEM findings indicate that household food waste is shaped more strongly by practical capabilities and everyday routines than by stated intentions or general food management norms. Importantly, the results demonstrate that food waste is not a uniform behaviour but consists of distinct waste pathways requiring differentiated policy and intervention responses.

From a sustainability perspective, these findings contribute directly to global efforts under the United Nations Sustainable Development Goals, particularly SDG 12 (Responsible Consumption and Production) and SDG 11 (Sustainable Cities and Communities). By differentiating between ingredient waste and cooked food waste, the study advances a more nuanced understanding of household-level responsible consumption and highlights how targeted behavioural levers can reduce inefficient resource use and prevent avoidable organic waste at the consumption stage.

Accordingly, food waste reduction policies should avoid one-size-fits-all approaches and instead adopt targeted, pathway-specific interventions. For ingredient-wasting households, interventions focused solely on planning, storage, or leftover management may be ineffective in highly urbanised contexts like Ho Chi Minh City. Instead, policies should prioritise strengthening cooking ability, improving portion adaptation skills, and enhancing alignment between purchased ingredients and actual household consumption needs through practical cooking education and flexible meal planning guidance.

For households that primarily waste cooked food, targeted interventions are needed to address mismatches between prepared meals and family food preferences. Strategies such as preference-sensitive meal planning, smaller batch cooking, and adaptive cooking practices may be particularly effective. Importantly, emotional appeals based on guilt should be applied with caution in this group, as higher levels of guilt were associated with increased food waste. Rather than moral or emotional messaging alone, interventions should prioritise enabling structures (such as improving cooking competence and household coordination) to support sustained behavioural change.

To complement the CB-SEM analysis, an exploratory Variance-Based SEM (VB-SEM) was conducted for the subgroup of zero-waste households. Unlike food-wasting households, this group provides insight into behavioural configurations associated with sustained waste prevention rather than waste generation.

A key distinction emerging from the VB-SEM results is the functional role of behavioural intention. In contrast to the CB-SEM models, intention in zero-waste households significantly predicted multiple downstream behaviours, including planning, storage practices, understanding family food preferences, and feelings of guilt. This indicates that among households that consistently avoid food waste, intention successfully translates into action. The absence of an intention–behaviour gap in this group highlights that

motivation can be effective when supported by adequate skills, routines, and household coordination.

Understanding family food preferences again plays a central role, both directly and indirectly influencing upstream food management behaviours. In zero-waste households, this capability predicts planning behaviours, which in turn predict storage practices. This layered structure suggests that accurate knowledge of household needs functions as a foundational organising mechanism that enables effective food management and prevents over-preparation. These findings extend the CB-SEM results by illustrating not only that this capability matters, but how it operates within a coherent behavioural system.

Cooking ability, while critical in food-wasting households, does not significantly predict planning or storage practices in the zero-waste group. This pattern suggests that cooking skills in zero-waste households may already be well-developed and internalised, functioning as a baseline capability rather than an active driver of behavioural variation. In this sense, cooking ability appears to be a necessary but non-differentiating condition once households reach a high level of food management performance.

The emotional role of guilt also differs markedly across groups. In zero-waste households, guilt is positively predicted by intention, indicating that it functions as a proactive moral commitment rather than a reactive emotional response. This contrasts sharply with the cooked-food-wasting group, where guilt is associated with higher waste levels. Together, these findings suggest that emotions such as guilt may either reinforce or undermine sustainable behaviour depending on whether households possess the practical capacity to act on moral intentions.

Overall, the combined CB-SEM and VB-SEM results demonstrate that effective food waste prevention emerges from the alignment of motivation, capability, and everyday practices. This distinction provides valuable insights for intervention design, suggesting that efforts to reduce household food waste should move beyond awareness-raising and instead focus on enabling households to operationalise intentions through skill-building, coordination, and context-appropriate food management strategies.

Concluding remarks

This study examined the behavioural drivers of household food waste during home cooking in Ho Chi Minh City, Vietnam, using a mixed-SEM approach grounded in the Theory

of Planned Behaviour and extended by capability- and practice-based factors. By combining CB-SEM and exploratory VB-SEM analyses, the study provides a differentiated understanding of household food waste across dominant waste patterns and highlights how behavioural mechanisms vary across household types.

The CB-SEM results show that household food waste is shaped by distinct behavioural pathways rather than a single, uniform structure. Across household groups, behavioural intention did not directly predict actual food waste, underscoring a persistent intention–behaviour gap. In contrast, capability-based factors (particularly cooking ability and understanding family food preferences) consistently reduced food waste across models. Emotional responses also played divergent roles across waste types, indicating that behavioural determinants cannot be assumed to operate uniformly across household contexts.

The exploratory VB-SEM analysis of zero-waste households offers complementary insights into the conditions under which food waste prevention becomes sustained. In this group, intention was effectively translated into planning and storage behaviours, supported by strong household coordination and internalised routines. Together, the findings suggest that effective food waste prevention depends not on motivation alone, but on the alignment between intention, capability, and everyday practices.

From a sustainability perspective, these findings reinforce the need to move beyond one-size-fits-all behavioural interventions. In highly urbanised contexts where food purchasing is frequent, interventions focused solely on planning or storage may be insufficient. Instead, policies should prioritise strengthening practical cooking competence, improving alignment between prepared meals and household needs, and supporting households in translating intentions into action. By differentiating between ingredient waste and cooked food waste, this study contributes to more targeted and context-sensitive approaches to reducing household food waste, advancing both responsible consumption (SDG 12) and sustainable urban systems (SDG 11).

Despite the above contributions, this study has some limitations. Household food waste was measured using self-reported 7-point Likert scale responses, capturing participants' perceptions rather than actual waste quantities. While this approach offers valuable insight into behavioural tendencies, it may not fully reflect the true volume of waste generated. Future research should consider complementing self-reported data with direct measurement methods (such

as kitchen waste audits or food diaries) to enhance data accuracy and validity.

Second, the data were collected between September and October 2023, offering a cross-sectional snapshot of household food waste behaviours at a single point in time. As food waste behaviours may evolve in response to changing socio-economic conditions and recent waste management regulations in Vietnam, this design limits insight into behavioural dynamics and temporal shifts. Future studies could therefore adopt longitudinal designs or latent transition analyses to examine how households move between different waste profiles (e.g., ingredient-wasting, cooked-food-wasting, and zero-waste) over time.

Finally, for reasons of analytical focus and survey practicality, this study concentrated on food waste related to meat, fruits, and vegetables, which constitute core components of

home-cooked meals in urban Vietnam. However, other food categories (such as rice, bread, dairy products, packaged foods, and condiments) also contribute to household food waste and play important roles in everyday cooking practices. Future research should therefore adopt a broader food categorisation to capture a more comprehensive picture of household food waste and its behavioural drivers.

Appendix A

Outputs of the data analysis process

See Fig. 5 and Table. 9,10,11,12,13,14,15 here

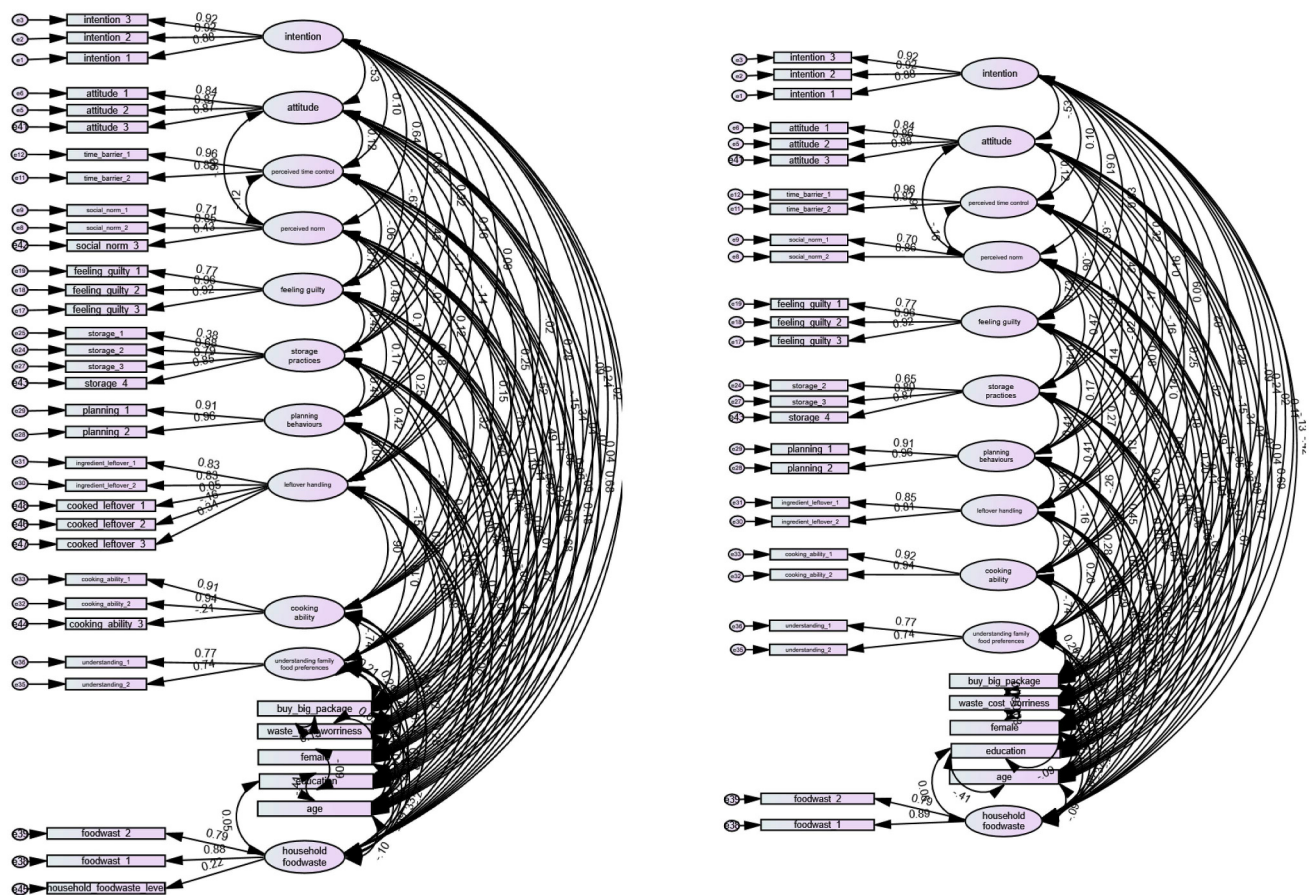


Fig. 5 Factor loadings of the Confirmatory Factor Analysis to test the measurement model (CB-SEM)

Table 9 The measurements and descriptive statistics of the variables/constructs

Variable/Constructs	Items	Measurements	Mean (Std. dev.)	Min	Max
Intention					
	intention_1	I am willing to reduce my household food waste in the future.	5.83 (1.17)	1	7
	intention_2	I intend to reduce my household food waste in the future.	5.74 (1.22)	1	7
	intention_3	I will conduct the best practices to reduce my household food waste levels.	5.8 (1.16)	2	7
Attitudes					
	attitude_1	Preference toward reducing household food waste.	6.24 (0.97)	1	7
	attitude_2	Reducing household food waste is useful.	6.3 (0.94)	1	7
	attitude_3	Reducing household food waste is wise.	6.25 (0.92)	2	7
Perceived norms					
	social_norm_1	Many important people to me pay attention to reducing household food waste.	5.89 (1.15)	1	7
	social_norm_2	Reducing household food waste is doing the right thing.	6.28 (0.91)	2	7
	social_norm_3	Households around me are trying to reduce their food waste.	5.17 (1.35)	1	7
Perceived time control					
	time_barrier_1	I have no time to conduct the appropriate food management at home.	3.25 (2.02)	1	7
	time_barrier_2	I have limited time when purchasing food at the shops.	3.21 (1.98)	1	7
Feeling guilty					
	feeling_guilty_1	I feel guilty when my household wastes food, which hurts the environment.	5.79 (1.26)	1	7
	feeling_guilty_2	I feel guilty when my household wastes food, as others do not have enough to eat.	6.01 (1.15)	2	7
	feeling_guilty_3	I feel guilty when my household wastes food, as “food waste is a sin”.	5.96 (1.17)	2	7
Storage practices					
	storage_1	After buying the ingredients, I prepare them once and store them in different food storage boxes until I purchase the next ingredient.	4.95 (1.80)	1	7
	storage_2	I set an appropriate storage order in the fridge to observe and use the food before it spoils or becomes outdated.	5.50 (1.42)	1	7
	storage_3	I regularly check my fridge and pantry to use the food before it spoils or becomes outdated.	5.47 (1.42)	1	7
	storage_4	Before shopping, I always check what food is in the fridge and pantry.	5.57 (1.34)	1	7
Planning behaviours					
	planning_1	Before going food shopping, I always plan the meals to be cooked.	5.19 (1.90)	1	7
	planning_2	I always plan what ingredients to buy using the shopping lists.	5.02 (1.92)	1	7
Leftover handling					
	ingredient_leftover_1	I always store the ingredient leftovers in appropriate conditions so that they will last and be used later	5.00 (1.83)	1	7
	ingredient_leftover_2 (rv)	<i>I save leftover ingredients in the fridge but throw them away afterwards.</i>	4.86 (1.89)	1	7
	cooked_leftover_1 (rv)	<i>When there is leftover cooked food, we always throw it away immediately.</i>	4.23 (2.12)	1	7
	cooked_leftover_2 (rv)	<i>I save leftover cooked food in the fridge but throw it away afterwards.</i>	4.76 (1.96)	1	7
	cooked_leftover_3	When there is leftover food, we use it to make new dishes.	3.74 (1.98)	1	7
Cooking ability					
	cooking_ability_1	I have high cooking skills.	2.54 (1.25)	1	7
	cooking_ability_2	I have high cooking experience.	2.49 (1.26)	1	7
	cooking_ability_3 (rv)	<i>I am not good at cooking at home.</i>	5.01 (1.92)	1	7

Table 9 (continued)

Variable/Constructs	Items	Measurements	Mean (Std. dev.)	Min	Max
Understand- ing family food preferences					
	understanding_1	I understand my family's food preferences.	5.95 (1.07)	1	7
	understanding_2	I prepare portions that are adequate for the actual needs of each family member.	6.04 (0.97)	1	7
Household food waste					
	foodwaste_1	We are trying to limit the amount of food waste at home.	1.75 (0.94)	1	6
	foodwaste_2	My household rarely wastes food that could have been avoided.	1.83 (1.03)	1	6
Buying big packages	-	I buy big packages, although they are too big for my household's needs.	4.27 (1.90)	1	7
Waste cost worriness	-	I am worried about the cost of food that I throw away.	5.58 (1.51)	1	7
Age	-	Age of the respondent.	51.38 (13.77)	19	82
Gender	-	0 = male (reference) 1 = female	Male: 22/74% Female: 77.26%		
Education	-	0 = never go to school 1 = primary school 2 = secondary school 3 = high school 4 = college 5 = undergraduate 6 = postgraduate	Never go to school: 1.4% Primary school: 18.38% Secondary school: 26.79% High school: 27.41% College: 6.7% Undergraduate: 15.89% Postgraduate: 3.43%		

All items were measured using a 7-point Likert scale. Underlined statements indicate reverse-coded items. Standard deviations are shown in parentheses. Descriptive statistics are based on responses from 642 households.

Table 10 Cronbach's Alpha values of the latent constructs (CB-SEM)

Latent variable	Number of items in the scale	Scale reliability coefficient (Cronbach's Alpha)
Intention	3	0.933
Attitude	3	0.897
Perceived norms	2	0.738
Perceived time control	2	0.884
Feeling guilty	3	0.909
Storage practices	3	0.811
Planning behaviours	2	0.935
Leftover handlings	2	0.813
Cooking ability	2	0.925
Understanding family food preferences	2	0.723
Household food waste level	2	0.822

Table 11 Discriminant validity HTMT (CB-SEM)

	Intention	Attitude	Per- ceived norms	Per- ceived time control	Feel- ing guilty	Storage practices	Planning behaviours	Leftover handling	Cook- ing ability	Under- standing family food preferences	House- hold food waste level
Intention	N/A										
Attitude	0.526	N/A									
Perceived norms	0.606	0.909	N/A								
Perceived time control	0.1	0.121	0.159	N/A							
Feeling guilty	0.635	0.631	0.735	0.054	N/A						
Storage practices	0.316	0.425	0.474	0.158	0.419	N/A					
Planning behaviours	0.154	0.166	0.136	0.016	0.173	0.412	N/A				
Leftover handling	0.098	0.165	0.212	0.076	0.267	0.414	0.099	N/A			
Cooking ability	0.018	0.253	0.317	0.139	0.13	0.256	0.156	0.065	N/A		
Understanding family food preferences	0.28	0.52	0.602	0.179	0.403	0.452	0.273	0.202	0.744	N/A	
Household food waste level	0.422	0.688	0.673	0.171	0.471	0.405	0.21	0.23	0.267	0.538	N/A

Table 12 Standardised Regression Weights of the General Household Food Waste Model using Percentile Bootstrap Confidence Intervals

Paths	General Household Food Waste Model ($R^2 = 0.71$)			
	Estimate (β)	95% CI Lower	95% CI Upper	p -value
Intention $\leftarrow$ Attitudes	1.992***	0.372	12.360	0.002
Intention $\leftarrow$ Perceived Time Controls	0.296***	0.153	0.741	0.005
Intention $\leftarrow$ Perceived Norms	2.626***	1.016	13.058	0.000
Household Food Waste $\leftarrow$ Intention	−0.008	−0.232	0.340	0.882
Household Food Waste $\leftarrow$ Perceived Time Controls	0.085	−0.094	0.271	0.330
Household Food Waste $\leftarrow$ Feeling Guilty	0.179	−0.101	0.604	0.202
Household Food Waste $\leftarrow$ Storage Practices	0.123	−0.168	0.482	0.394
Household Food Waste $\leftarrow$ Planning Behaviours	0.099	−0.077	0.412	0.298
Household Food Waste $\leftarrow$ Leftover Handlings	0.028	−0.172	0.313	0.797
Household Food Waste $\leftarrow$ Cooking Ability	−1.244***	−2.551	−0.618	0.000
Household Food Waste $\leftarrow$ Understanding Family Food Preferences	−1.831***	−3.430	−1.113	0.000
Household Food Waste $\leftarrow$ Waste Cost Concerns	−0.180*	−0.498	−0.009	0.063
Household Food Waste $\leftarrow$ Buying Big Packages	0.211*	−0.017	0.607	0.072
Household Food Waste $\leftarrow$ Female	0.066	−0.089	0.292	0.471
Household Food Waste $\leftarrow$ Education	0.112	−0.046	0.289	0.156
Household Food Waste $\leftarrow$ Age	−0.050	−0.239	0.119	0.531

Notes: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$

Table 13 Distribution of ingredients and cooked food waste shares by household waste group

Waste group	Waste type	Mean (%)	Median (%)	SD (%)	Min (%)	Max (%)	N
Group 1	Ingredient waste	83.7	90	16.2	60	100	246
	Cooked food waste	16.3	10	16.2	0	40	246
Group 2	Ingredient waste	18.6	20	14.9	0	45	173
	Cooked food waste	81.4	80	14.9	55	100	173
Total sample	Ingredient waste	56.8	60	35.7	0	100	419
	Cooked food waste	43.2	40	35.7	0	100	419

Percentages are based on respondents' estimates of the share of total household food waste attributable to raw ingredients and cooked food, assuming total food waste equals 100%. Waste groups are defined by the dominant waste share (ingredient vs. cooked food), reflecting predominant rather than exclusive waste patterns. Households reporting an equal split between ingredient and cooked food waste (50–50) were excluded from this classification to maintain analytical focus on contrasting dominant waste profiles.

Table 14 Initial assessment of construct validity and reliability (VB-SEM)

Constructs	Cronbach's Alpha (threshold: ≥ 0.70)	CR (threshold: ≥ 0.70)	AVE (threshold: ≥ 0.70)
Attitude	0.942	0.963	0.897
Cooking ability	0.610	0.802	0.625
cooking_ability_1	0.962	-	-
cooking_ability_2	0.966	-	-
cooking_ability_3	0.130	-	-
Feeling guilty	0.963	0.976	0.931
Intention	0.962	0.975	0.929
Leftover handling	0.369	0.542	0.464
Perceived norms	0.848	0.909	0.772
Perceived time controls	0.936	0.968	0.939
Planning behaviours	0.953	0.977	0.955
Storage practices	0.885	0.922	0.749
Understanding family food preferences	0.869	0.937	0.882

Table 15 Discriminant validity HTMT (VB-SEM)

	Age	Attitude	Cook- ing ability	Education	Feel- ing guilty	Gender	Intention	Per- ceived norms	Perceived time controls	Planning behaviours	Storage practices	Under- standing family food preferences
Age	N/A											
Attitude	0.047	N/A										
Cooking ability	0.145	0.225	N/A									
Education	0.353	0.185	0.035	N/A								
Feeling guilty	0.047	0.780	0.306	0.256	N/A							
Gender	0.109	0.046	0.123	0.119	0.048	N/A						
Intention	0.051	0.773	0.306	0.242	0.930	0.030	N/A					
Perceived norms	0.099	0.860	0.477	0.228	0.869	0.039	0.900	N/A				
Perceived time controls	0.091	0.048	0.230	0.026	0.056	0.130	0.048	0.120	N/A			
Planning behaviours	0.130	0.288	0.186	0.344	0.384	0.164	0.390	0.354	0.144	N/A		
Storage practices	0.122	0.446	0.342	0.333	0.687	0.093	0.698	0.584	0.103	0.537	N/A	
Under- standing family food preferences	0.036	0.501	0.660	0.262	0.529	0.181	0.528	0.644	0.041	0.430	0.491	N/A

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Declarations

Conflict of interest The authors have no competing interests to declare that are relevant to the content of this article.

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